PRISM AIR Submission Information

PRISM AIR (Antibody Internalization Reporter) is a unique assay within APS screens for antibody (unconjugated) test agents only. Antibodies are coupled with a cytotoxic payload via a secondary drug-conjugate prior to screening. AIR screens are run in the same format and at the same time as APS screens: 5-day assay, 900+ cell lines, test agents run at 8 pt. dose for single-agents, with 3-fold dilutions in triplicate. Collaborators select the top screening dose as a part of the submission process, but there is a ***maximum top dose*** of 2 μg/mL due to constraints with the secondary drug-conjugate. For additional information about this assay, please see the PRISM AIR Overview slide deck or visit the PRISM website

Test Agent Requirements

- We require at least 500uL of 500X the top screening dose in an aqueous solution.
    - i.e. If your top screening dose is 2 μg/mL, you will submit a 1 mg/mL stock.
- Antibodies must be IgG isotype and contain a human or humanized Fc region
    - The secondary fab-drug conjugate is specifically designed to bind only to human IgG Fc regions. Antibodies lacking an Fc region or with non-human Fc regions are not compatible with the AIR assay.
- The maximum top screening dose for AIR test agents is 2 μg/mL
- The collaborator is responsible for ensuring that the test agent is submitted with accurate solvent information and for QC'ing test agents ahead of submission.
    - We do not have the bandwidth to do any optimization for, or re-running of, test agents that do not perform well in PRISM. Any test agents submitted will be run 'at-risk'
    - We are unable to solubilize powdered stock of compounds for collaborators.
- Liquid handling on our automation instruments has been validated using antibodies, ADCs, and cytokines. There is no guarantee that our liquid handlers will work with every kind of molecule or solution. Please refer to the list of labware below for reference.

Compound Handling Workflow

NOTE *Lab work for PRISM screens (from compound submission to lysate generation) can take up to 2 months and compounds may go through multiple freeze/thaw cycles. It is important that compounds submitted for consortium screens be tested for stability and QCed prior to submission.*

1. Test agents are shipped or delivered to the Broad Institute prior to the submission deadline. They are stored according to the conditions specified on the collaborator submission form until step 2 begins (-20°C, 4°C or room temperature (RT)).
2. After the submission window has closed, test agents will be transferred to the proper automation-friendly vials (catalog numbers listed below). These vials may be stored at 4°C for up to 1 week after transfer (dependent on when source plates are created).
    a. **For PRISM AIR,** antibodies submitted by collaborators are first mixed 1:1 (by volume) with the PRISM Fab-drug conjugate (FDC) reagent solution before being transferred to these vials. This

Updated: June 2026

is why AIR stock solutions must be submitted at 500x, instead of the 250x required for APS stock solutions. See details at the end of this document for more information about the FDC.

3. The test agents are added to source plates and diluted using a Hamilton liquid handler (several hours at RT). Test agents are cherry-picked from source vials into microplates and then serially diluted in H2O. Source dilution plates are sealed and may be stored at 4°C for up to 24 hours. 2 copies of source plates are created.
4. Pooled PRISM cells are plated into empty microplates (40uL) and incubated until Echo treatment.
    a. Adherent Lines- RPMI-10% FBS 1250 cells/well
    b. Mixed Lines- RPMI-20% FBS, 2000 cells/well
    c. Suspension Lines- RPMI-20% FBS, 3000 cells/well
5. Source dilution plates are transferred acoustically to cell plates (160nL from plate is added to 40uL cell media) using an Echo liquid handler. Cells are incubated for 5 days before being lysed at the endpoint.

Important Technical Notes for all PRISM aqueous screens

1. Liquid handling and cell treatments have been validated using H2O. This is the diluent used for creating dosed dilution plates. Due to the consortium format of our screens, we cannot accommodate any custom or specialized diluents for individual test agents. We advise you to confirm your test agent is soluble in H2O. Our data processing pipeline also uses 1 vehicle control to normalize all data (the H2O control) regardless of the stock solution submitted.
    a. Though we accept stock solutions of any aqueous formulation, these solutions will be diluted with H2O during the serial dilution process for all subsequent doses. Consider this for your top dose, which will not be diluted in H2O at all (direct transfer from stock solution vial to top dose wells).
    b. As the total assay duration is 5 days, please take into consideration the stability of your test agents in cell culture media at RT/37°C.
2. Liquid handling on our automation instruments have been validated using antibodies, ADCs, and cytokines. There is no guarantee that our liquid handlers will work with every kind of molecule or solution. Our instrument keeps a log of all wells unable to be transferred via Echo (often due to solubility issues).
    a. While we are able to screen cytokines in our aqueous workflow, our assay is only set up to measure growth inhibition at the moment. We do not measure growth-promoting effects of cytokines.
3. Please refer to the list below of labware used in our compound plating workflow for reference:

Item Description	Vendor	Catalog #
TV vials	Tradewinds	HRGV1545C-PPW13425
Matrix™ 2D Barcoded Clear Polypropylene Open-Top Storage Tubes. 0.5ml V bottom non-sterile	Thermo Scientific	3734
Microcentrifuge tube, 2mL, clear, screw-cap	VWR	16466-060
Axygen® 384-well tips, 30µL, Clear, Non-filtered, Non-sterile, SLAS Rack	Corning	FX-384-R
50 µL Nested Conductive Tips, Non-Filter, Non-Sterile,	Hamilton	235947

Nested Tip Racks, Black Polypropylene Tip		
384-well Polypropylene (PP) microplate, clear	Labcyte	PP-0200
384-well Low Flange White Flat Bottom Polystyrene TC-treated Microplates	Corning	3570

Important Technical Notes for PRISM AIR

1. Antibodies will be mixed by the PRISM team with a secondary Fab-drug conjugate (FDC) provided by PRISM
    a. The FDC is added at a fixed 1:2 Ab:FDC concentration ratio by weight (~1:6 molar ratio). To achieve this concentration ratio, the FDC reagent is prepared at double the concentration of the antibody stock solution. Next, equal volumes of the antibody stock and FDC are combined. This produces a ready-to-use solution where the antibody is now at 250x concentration.
    b. For example, if the desired final treatment concentration is 1 ug/mL, collaborators should submit a 500 ug/mL antibody solution. The PRISM team will prepare a 1000 ug/mL FDC solution. The final mixture is prepared by adding an equal volume of the antibody stock to the FDC solution.
2. The FDC is engineered to bind to the Fc region of a primary antibody. At this time, only one FDC option is available. Its components are:
    a. Antibody: A monoclonal anti-human IgG(Fc) Fab fragment
    b. Linker: A non-cleavable NHS linker
    c. Payload: monomethyl auristatin F (MMAF)
3. **Antibodies must be IgG isotype and contain a human or humanized Fc region**
    a. The secondary FDC is specifically designed to bind only to human IgG Fc regions.
    b. Antibodies lacking an Fc region or with non-human Fc regions are not compatible with the AIR assay.
4. **The maximum top screening dose for antibodies in PRISM AIR is 2 ug/mL.**